

PRINCE OMUYEH

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EDUCATION

Villanova University

Master of Science in Software Engineering

Villanova, PA

August 2026 – Present

Lehigh University, P.C. Rossin College of Engineering

Bachelor of Science in Computer Science, Minor in Data Science

Bethlehem, PA

August 2022 - May 2026

Relevant Coursework: Data Structures & Algorithms, Operating Systems Design, Distributed Systems, Database Systems, Computer Architecture, Systems Software, AI Theory & Practice, Data Mining

SKILLS

Languages: Python, Java, JavaScript, SQL, C++, R, Rust, HTML/CSS

Frameworks & Libraries: React, Node.js, Supabase, Spring, Maven

Developer Tools: Git, Docker, Linux, Figma, PostgreSQL, Jira/Atlassian, Agile/Scrum

EXPERIENCE

Computer Vision Engineer

Hawkeye (Capstone Sponsor)

January 2025 – December 2025

Bethlehem, PA

- Collaborated with startup founders to scope and deliver a real-time cyclist hazard detection pipeline, translating product requirements into engineering deliverables on a 4-person Agile team
- Migrated the object detection model from YOLOv8 to YOLOv11 by updating model architecture configs, refactoring preprocessing and postprocessing logic, and resolving dependency changes across the Ultralytics codebase
- Achieved a vehicle orientation prediction accuracy of 92.66%

Software Development Intern

Lehigh University

March 2025 – May 2025

Bethlehem, PA

- Owned end-to-end delivery of an admin interface for an academic integrity platform, enabling 100+ faculty and associate chairs to manage user roles and access student reports
- Implemented 6 RESTful APIs and OAuth authentication supporting role-based access control for faculty, associate chairs, and student accounts

PROJECTS

Multiplayer Game Systems Development | LuaU, Roblox Studio, Trello

January 2024 – Present

- Led product planning and sprint execution for a 6-person team, defining feature roadmaps, triaging community feedback, and coordinating releases for a 2,000+ member player base
- Designed and iterated on interactive UI/UX and reusable gameplay systems based on community feedback, optimizing rendering and physics logic for performance and user experience
- Managed bug and feature request tracking using BetaHub, maintaining product quality across active development cycles

Movie Watchlist Web Application | React, Node.js, PostgreSQL

April 2026 – May 2026

- Led front-end development for a full-stack movie watchlist app with a team of 3, delivering movie search, personal watchlists, and social reviews powered by the TMDb API
- Implemented a complete authentication system with email/password login, Google OAuth, and JWT session management using Supabase Auth
- Produced technical documentation covering system requirements, API contracts, and architecture decisions

Distributed Replicated Hash Table | C++

April 2026

- Implemented a distributed hash table across 5 nodes with Paxos-based consensus, supporting atomic operations, data replication, and fault tolerance
- Achieved ~30,000 ops/sec throughput and sub-1ms average latency at 10k keys under fault injection testing, maintaining availability after node crashes with no manual intervention

JPMorgan Chase Transaction Processing System | Java, Spring, Kafka, SQL

April 2025

- Designed a Kafka-based transaction processing system in Spring Boot backed by an H2 SQL database
- Developed transaction pipelines, external API interactions, and user balance endpoints using Java, Maven, and Spring